

KOA MARIE GERVAIS NELLY

Lead AI Engineer & Data Architect | Founder @ Archi Cam AI

EXECUTIVE SUMMARY

Lead AI Engineer & Data Architect (Google Developer Program Member), leveraging Google Antigravity to design neuro-symbolic multi-agent architectures, Neo4j GraphRAG, and sovereign MLOps pipelines. Expert in AI ethics: SHAP transparency, EU AI Act compliance, and deterministic zero-hallucination models. Founder of Archi Cam AI (official Google Africa Applied AI Lab applicant 2026), bridging civil engineering methodology, mathematical rigor, and product vision.

FLAGSHIP AI PROJECTS

Archi Cam AI — *Agentic AI SaaS & 5D BIM*

- Official Applicant Google Africa Applied AI Lab. Built with Google Antigravity (Gemma 4 12B, Gemini 2.5, BAEI 91).
- Automated Excel BOQs in <45s (−99.2% time, MLflow $R^2=0.9872$) and 3D renders via Imagen 3 + ControlNet.

K1-MATHINFO (v3.0.0) — *Sovereign Multi-Agent AI, WikiSkills & OKF Certification*

- Sovereign DMI system (Univ. of Ngaoundéré): 470 theses, 19 M1 projects, 1,366 Neo4j nodes.
- 6 LangGraph agents orchestrated via WikiSkills (critic anti-hallucination, Cypher optimizer), RRF k=60, SHA-256 No-LLM (100%).

Sovereign.BI Agentic — *Enterprise Security & Agentic BI*

- NL-to-SQL/Graph engine (PostgreSQL pgvector, Neo4j N10S, <5s latency) with ABAC guardrails + SHAP Sentinel.

Dataset Automator & VigieSahel — *MLOps Pipeline & Climate AI Impact*

- Dataset Automator: Google Antigravity (Google Cloud Hackathon) — TabFM, BigQuery DataFrames, EU AI Act.
- VigieSahel: −35% crop loss, +14d epidemic forecast (XGBoost $R^2>94\%$, Supabase, MLflow).

PROFESSIONAL EXPERIENCE

AI Lead & Data Science Consultant — 2025 – Present

Independent Projects & Enterprises | Douala, CM

- Ethical sovereign AI systems: GraphRAG Neo4j pipelines, high-dimensional EDA, multi-source RAG architectures and executive KPI dashboards.

Aviation Security Officer (AVSEC) — 2018 – Present

CCAA — Cameroon Civil Aviation Authority | Douala, CM

- Threat assessment, secure access control, and regulatory compliance audits (ICAO Annex 17).

EDUCATION & CERTIFICATIONS

M.Sc. in Applied Artificial Intelligence — 2025 – 2027 [In Progress]

University of Ngaoundéré | Cameroon

- ML & Bayesian Statistics, Data Engineering & Neo4j, Computer Vision & Robotics, Ethics & Cybersecurity, Production MLOps.

B.Sc. in Civil Engineering (Building Option) — 2015 – 2016

ISTDI / IUC Douala | Cameroon

- Structural calculations (BAEI 91), quantity surveying, construction project management — AI-applied estimation base.

HONORS & APPLIED AI DISTINCTIONS

- Google Africa Applied AI Lab (Accra, 2026):** Official candidacy — Archi Cam AI platform (5D BIM + GenAI).
- Google Cloud #AllThingsAgentic Hackathon:** Dataset Automator v4.0 (Google Antigravity, TabFM, bigframes, WIT, MCT).



LEAD AI ENGINEER

Founder · Archi Cam AI

CONTACT & PROFILES

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△ Douala / Ngaoundéré, CM

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📄 linkedin.com/in/marie-gervais-koa

📧 devpost.com/magenel85

📄 Google Developer Program

AI & LLM STACK

Google Antigravity IDE ■■■■■■

LangGraph & WikiSkills ■■■■■■

Google Gemma 4 (12B) ■■■■■■

Gemini 2.5 / 1.5 Pro ■■■■■■

Google TabFM (Tabular) ■■■■■■

Firebase Genkit ■■■■■■

Neo4j GraphRAG — Agent K1

DATA & GRAPHS

Neo4j / Cypher Graph ■■■■■■

PostgreSQL / pgvector ■■■■■■

Google BigQuery ■■■■■■

Supabase Realtime ■■■■■■

Pandas / NumPy ETL ■■■■■■

DEV & MLOPS

Python 3.11+ / MLOps ■■■■■■

FastAPI / Next.js 14 ■■■■■■

MLflow & Data Drift ■■■■■■

SHAP Sentinel Audit ■■■■■■

Docker & Streamlit ■■■■■■

IfcOpenShell (5D BIM) ■■■■■■

ETHICS, SECURITY & AUDIT

- AI Ethics & Anti-Hallucination
- Quorum 4-Eyes Governance
- OKF v0.2 SHA-256 No-LLM
- EU AI Act Compliance (RSASSA)
- AVSEC (ICAO Annex 17)

LANGUAGES

French — Native / Fluent

English — Functional basics (tools & doc.)

KEY ASSETS

- Dual Competence: AI & Civil Eng.
- Aviation Security & Crisis (CCAA)
- Mathematical Rigor & Guardrails